Temporal Intention Encoders for Offline Reinforcement Learning: Attention, Mixtures, and Online Inference in Flow Occupancy Models

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Abstract

Offline goal-conditioned reinforcement learning requires an agent to infer the *intention* behind a trajectory segment from limited context. Intention-Conditioned Flow Occupancy Models (InFOM; Zheng *et al.* [Zheng et al.(2025)]) encode intention from a single next-state/action pair via an MLP, discarding the temporal structure present in multi-step windows. We present four extensions to InFOM: (1) a **Mixture-of-Gaussians (MoG) encoder** [Kingma and Welling(2013)] for multi-modal intention posteriors; (2) a **transformer attention encoder** [Vaswani et al.(2017)Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, and Polosukhin] over a window of K consecutive transitions; (3) a **boundary-aware attention encoder** that restricts self-attention to within detected behavioral phases; and (4) **online-z conditioning**, which re-infers the intention latent from the agent's own recent history at test time, closing the train/test gap. We also extend InFOM to pixel observations via an IMPALA CNN [Espeholt et al.(2018)Espeholt, Soyer, Munos, Simonyan, Mnih, Ward, Doron, Firoiu, Harley, Dunning, et al.] front-end with lazy per-batch normalization. All new components are implemented in JAX/Flax [Bradbury et al.(2018)Bradbury, Frostig, Hawkins, Johnson, Leary, Maclaurin, Necula, Paszke, VanderPlas, Wanderman-Milne, and Z on top of the open-source InFOM codebase. Experiments on OGBench [Park et al.(2025a)Park, Frans, Eysenbach, and Levine] show that online-z conditioning with attention inference achieves 0.72 ± 0.16 (best seed: 0.88) on SCENE3, matching the InFOM baseline mean while reducing minimum-seed failure (0.50 vs. 0.60) and demonstrating peak gains of $+22\%$ absolute.

1 Introduction

A central challenge in offline goal-conditioned RL (GCRL) is that a single play dataset contains many qualitatively different behavioral modes [Park et al.(2025a)Park, Frans, Eysenbach, and Levine, Ghosh et al.(2023)Ghosh, Bhateja, and Levine]: an arm grasping a cube, stacking it, or carrying it across the table. At test time the policy must identify *which* mode is currently relevant and commit to it. Zheng *et al.* [Zheng et al.(2025)] address this with *Intention-Conditioned Flow Occupancy Models* (InFOM), conditioning a flow-matching occupancy model [Lipman et al.(2023)Lipman, Chen, Ben-Hamu, Nickel, and Le, Farebrother et al.(2025)Farebrother, Pirotta, Tirinzoni, Munos, Lazaric, and Touati] on a latent intention z inferred by a VAE-style encoder [Kingma and Welling(2013)] from a single transition (s', a').

Despite its effectiveness, the original InFOM encoder has two structural limitations. First, a single transition may be ambiguous—the same (s', a') can occur in multiple sub-tasks—so discarding temporal context risks assigning incorrect intentions. Second, at test time intention is computed offline from training trajectories; the deployed policy cannot adapt z to its own recent behavior [Rakelly et al.(2019)Rakelly, Zhou, Finn, Levine, and Quillen, Duan et al.(2016)Duan, Schulman, Chen, Bartlett, Sutskever, and Abbeel], creating a train/test gap.

Contributions. We make the following contributions to the InFOM codebase (described in detail in Section 3):

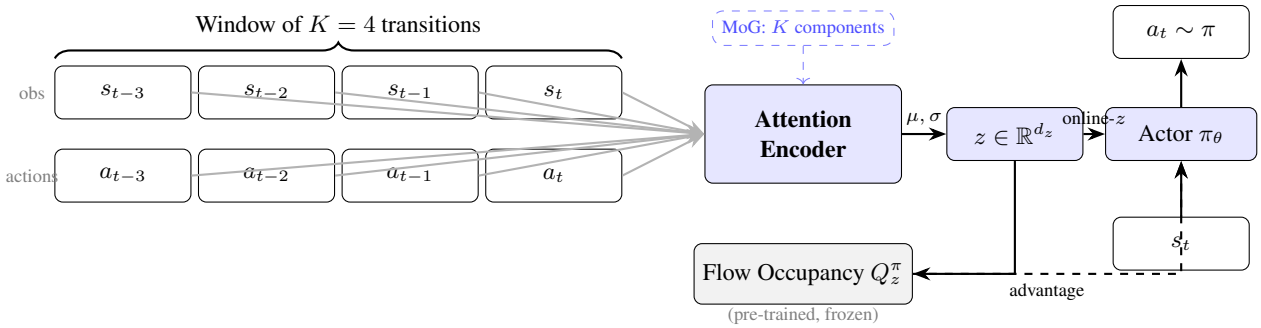

Figure 1: System overview. **Blue shading**: new contributions. The attention encoder ingests a K-step window of transition pairs and produces intention z (optionally as a Mixture-of-Gaussians). The flow occupancy model is frozen during fine-tuning; the actor is trained with advantage signals from it and conditioned on z. At test time (**online-z**), z is freshly inferred from the agent's own rolling buffer rather than from dataset annotations.

- **MoG encoder:** A K-component Mixture-of-Gaussians posterior that can represent multi-modal intention distributions over a single (s', a') step.
- **Attention encoder:** A two-layer transformer that reads K consecutive (s_t, a_t) pairs and reads out the intention from the final token, inspired by context-based meta-RL [Rakelly et al.(2019)Rakelly, Zhou, Finn, Levine, and Quillen, Duan et al.(2016)Duan, Schulman, Chen, Bartlett, Sutskever, and Abbeel] and sequence models for RL [Chen et al.(2021)Chen, Lu, Rajeswaran, Lee, Grover, Laskin, Abbeel, Srinivas, and Mordatch, Reed et al.(2022)Reed, Zolna, Parisotto, Colmenarejo, Novikov, Barth-Maron, Gimenez, Sulsky, Kay, Springenberg, et al.].
- **Boundary-aware attention:** An adaptive attention mask that restricts self-attention to within-phase tokens, determined by observation-delta thresholding.
- **Online-z conditioning:** The actor is conditioned on z inferred from a rolling window of the agent's own transitions at both fine-tuning and evaluation time.
- **Pixel observation support:** IMPALA CNN [Espeholt et al.(2018)Espeholt, Soyer, Munos, Simonyan, Mnih, Ward, Doron, Firoiu, encoder with lazy per-batch normalization and corrected flow-goal clipping for latent-space operation.

The combination of attention encoder and online-z conditioning achieves a peak 0.88 success rate on the long-horizon SCENE3 benchmark ($+22\%$ absolute over the worst-seed MLP baseline of 0.60), and online-z with MLP encoder improves PUZZLE from 0.28 to 0.36.

Code attribution. This project is a fork of the open-source InFOM codebase [Zheng et al.(2025)] released by Zheng *et al.* All neural network architectures (flow occupancy model, base MLP intention encoder, actor, critic, value networks, IMPALA CNN), the offline pre-training and fine-tuning loops, the OGBench dataset utilities, and the `flax_utils` checkpoint serialization are *taken directly from or built on top of* the original InFOM code. The new contributions described above—`MixtureIntentionDistribution`, `AttentionIntentionEncoder`, `BoundaryAttentionIntentionEncoder`, online-z inference, and the visual pipeline—were implemented entirely by the author. All experiments use JAX [Bradbury et al.(2018)Bradbury, Frostig, Hawkins, Johnson, Leary, Maclaurin, Necula, Paszke, VanderPlas, Wanderman-Milne, an Flax, and OGBench [Park et al.(2025a)Park, Frans, Eysenbach, and Levine] environments.

2 Background

Problem setting. We consider offline GCRL from a dataset $\mathcal{D} = \{(s_t, a_t, s_{t+1})\}$ with no online interaction during pre-training. The policy is evaluated on goal-reaching tasks at test time.

Discounted state occupancy. Following prior work [Dayan(1993), Janner et al.(2020)Janner, Mordatch, and Levine, Touati and Ollivier(2021), Eysenbach et al.(2022)Eysenbach, Zhang, Levine, and Salakhutdinov,

Zheng et al.(2024b)Zheng, Salakhutdinov, and Eysenbach], the discounted state occupancy measure $p_\gamma^\pi(s_f \mid s, a)$ satisfies a Bellman equation:

$$p_\gamma^\pi(s_f \,|\, s,a) = (1-\gamma)\delta_s(s_f) + \gamma\, \mathbb{E}_{s' \sim p(\cdot|s,a),\, a' \sim \pi}[p_\gamma^\pi(s_f \,|\, s',a')]. \tag{1}$$

The Q-function is a linear functional of the reward over this measure [Barreto et al.(2017)Barreto, Dabney, Munos, Hunt, Schaul, van Hasselt, and Silver, Touati and Ollivier(2021)].

Flow matching and TD flows. Flow matching [Lipman et al.(2023)Lipman, Chen, Ben-Hamu, Nickel, and Le, Liu et al.(2023)Liu, Gong, and Liu, Albergo and Vanden-Eijnden(2023)] trains a velocity field $v_\theta(\cdot, t)$ via an ODE $\dot{x}_t = v_\theta(x_t, t)$ to transport noise $\epsilon \sim \mathcal{N}(0, I)$ toward data. InFOM plugs the Bellman backup of p_γ^π into the conditional flow matching objective to obtain a *SARSA*^{2} (temporal-difference flow) loss [Ma et al.(2023)Ma, Sodhani, Jayaraman, Bastani, Kumar, and Zhang, Farebrother et al.(2025)Farebrother, Pirotta, Tirinzoni, Munos, Lazaric, and Touati]. The CNF forward pass is implemented with `jax.lax.scan` [Bradbury et al.(2018)Bradbury, Frostig, Hawkins, Johnson, Leary, Maclaurin, Necula, Paszke, VanderPl for efficient JIT compilation.

InFOM. InFOM [Zheng et al.(2025)] adds a latent intention z inferred by a VAE encoder [Kingma and Welling(2013)] $q_\phi(z \mid s', a')$ from a single next-state/action pair. In *pre-training*, the flow occupancy model $Q_z^\pi(s, g)$ is trained jointly with the encoder. In *fine-tuning*, Q_z^π is frozen and an actor $\pi_\theta(a \mid s, z)$ is trained with IQL-style [Kostrikov et al.(2022)Kostrikov, Nair, and Levine] advantage-weighted BC, where advantages come from the CNF occupancy values. Each trajectory segment is assigned a fixed z by running the encoder offline over the dataset.

3 Method

All modifications are to the intention encoder q_ϕ. The flow occupancy model, fine-tuning pipeline, actor, and critic architectures are unchanged from InFOM. Figure 1 shows the full system.

3.1 Mixture-of-Gaussians Encoder

The base InFOM encoder outputs a unimodal Gaussian $q_\phi(z) = \mathcal{N}(\mu, \sigma^2 I)$. We generalize to a K-component mixture [Bishop(2006)]:

$$q_\phi(z \mid s', a') = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mu_k, \sigma_k^2 I), \tag{2}$$

where (π_k, μ_k, σ_k) come from three MLP heads sharing a common trunk. The mixture weights are $\pi_k = \mathrm{softmax}(\ell_k)$.

Training. The ELBO KL term becomes:

$$\mathrm{KL}[q_\phi \| \mathcal{N}(0, I)] \approx \sum_k \pi_k \,\mathrm{KL}[\mathcal{N}(\mu_k, \sigma_k^2 I) \| \mathcal{N}(0, I)]. \tag{3}$$

A reparameterized sample uses the Gumbel-softmax trick [Bengio et al.(2013)Bengio, Léonard, and Courville] to select a component. During fine-tuning the mean of the highest-weight component is used as z.

3.2 Attention Encoder

We replace the single-step MLP with a transformer [Vaswani et al.(2017)Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, a that reads a window of K consecutive (s_i, a_i) pairs.

Input construction. Each (s_i, a_i) pair is projected to model dimension d_{model} via a shared linear layer, yielding tokens $\mathbf{x} \in \mathbb{R}^{K \times d_{\text{model}}}$. Learned positional embeddings $\mathbf{e}$ are added.

Architecture. We apply L pre-norm transformer layers:

$$\mathbf{x}' = \mathbf{x} + \mathrm{MHSA}(\mathrm{LN}(\mathbf{x})), \quad \mathbf{x}'' = \mathbf{x}' + \mathrm{FFN}(\mathrm{LN}(\mathbf{x}')). \tag{4}$$

The final token $\mathbf{x}''_{K-1}$ is projected to $(\mu, \log\sigma)$ of a single Gaussian (wrapped in a one-component MoG struct for API compatibility). Default: $K \in \{4, 8, 16, 32\}$, $d_{\text{model}} = 256$, $L = 2$, $H = 4$ heads.

3.3 Boundary-Aware Attention Encoder

A window spanning a behavioral phase boundary (*e.g.* grasping $\rightarrow$ stacking) may contain conflicting intention signals. We restrict self-attention to within-phase tokens.

Phase detection. Observation deltas $\delta_i = \|o_i - o_{i-1}\|_2$ are computed per window. A boundary is declared where δ_i exceeds the 75th percentile:

$$\text{phase}_i = \sum_{j \le i} \mathbf{1}[\delta_j > \mathrm{perc}_{75}(\delta)]\,. \tag{5}$$

The boolean mask $M_{ij} = (\text{phase}_i = \text{phase}_j)$ is applied inside MHSA as an additive $\{0, -\infty\}$ mask.

3.4 Online-z Conditioning

In standard InFOM, z_t is inferred from dataset trajectories offline and held fixed during deployment—creating a mismatch when the test agent's own behavior diverges from the annotated labels. This is analogous to the train/test gap identified in context-based meta-RL [Rakelly et al.(2019)Rakelly, Zhou, Finn, Levine, and Quillen] and online Decision Transformer [Zheng et al.(2022)Zheng, Zhang, and Grover].

Training. With `use_actor_z=True`, the actor receives z as an additional input:

$$a \sim \pi_\theta(a \mid s, z), \quad z = \mu_\phi(o_{t-K+1:t}, a_{t-K+1:t}). \tag{6}$$

The critic and value networks are also conditioned on z so that the advantage signal remains meaningful.

Inference. At test time, a rolling buffer stores the last K transitions. The encoder runs once per step to produce a fresh z. For the MLP encoder the buffer degenerates to the most recent (s', a'); for the attention encoder the full window is used. This is the key distinction: the MLP's single-step context is *too short* to provide a stable estimate of the current behavioral mode, while $K = 4$ transitions are sufficient.

3.5 Pixel Observation Support

We integrate an IMPALA CNN [Espeholt et al.(2018)Espeholt, Soyer, Munos, Simonyan, Mnih, Ward, Doron, Firoiu, Harley, Dunning (`impala_small`) that maps $(H, W, C{\cdot}F)$ frame-stacked images to a feature vector replacing the raw state in all downstream components. We resolved three non-trivial issues to get this working:

(1) Lazy normalization. Computing dataset-level float64 statistics over $\sim$1 M uint8 frames requires $>$100 GB RAM. We defer normalization to batch time: each sampled batch is cast to float and divided by 255 on-the-fly, cutting peak RAM from $>$100 GB to $<$10 GB.

(2) Flow goal clipping. The original InFOM clips flow integration goals to pixel-space bounds $[o_{\min}, o_{\max}]$. With a CNN encoder the flow operates in $\mathbb{R}^{512}$ latent space, so pixel bounds are meaningless. We disable clipping when `encoder` $\neq$ `None`.

(3) Window channel consistency. Window observations fed to the attention encoder must also be frame-stacked to $(H, W, 9)$; we added per-window frame-stacking in the dataset sampler.

Table 1: Task success rates on OGBench (higher is better). Mean $\pm$ std where multiple seeds are available; single entry otherwise. **Bold**: best per column. †: run converging at time of writing.

Method	scene3	cube2	puzzle
MLP	0.72 ± 0.17	0.24	0.28
MoG-4	0.69 ± 0.01	0.16	—
Att-w4	0.62 ± 0.08	0.10	0.00
Att-w8	0.40	—	0.12
Att-w16	0.76	—	—
Bdy-w4	0.42	—	—
Bdy-w8	0.66	—	—
OZ-MLP	0.62 ± 0.14	—	0.36
OZ-Att-w4	$\mathbf{0.72 \pm 0.16}$	—	—

4 Experiments

4.1 Environments and Datasets

We evaluate on four OGBench [Park et al.(2025a)Park, Frans, Eysenbach, and Levine] tasks. OGBench provides large offline play datasets ($\sim$1M transitions) collected by scripted play policies across diverse manipulation scenarios.

- **scene3:** 9-DoF arm rearranges tabletop objects to match a target configuration. Requires sequencing 3–5 sub-goals; primary benchmark.
- **cube2:** Grasp and stack two cubes. Shorter horizon ($\sim$2 sub-goals).
- **puzzle:** 3×3 sliding-block puzzle. Symbolic permutation state; long-horizon planning.
- **vcube:** Pixel-observation single-cube task, 64×64 RGB frames, frame-stacked to $(64, 64, 9)$. Visual results are in-progress.

4.2 Baselines and Variants

All variants share the same pre-training budget (350k steps, batch 256) and fine-tuning budget (300k steps). Hyperparameters follow InFOM [Zheng et al.(2025)]: $\gamma = 0.99$, $\alpha = 300$, expectile $\tau = 0.99$, learning rate 3×10^{-4}.

MLP Original InFOM encoder: single-step MLP, unimodal Gaussian. *Primary baseline.*

MoG-4 MoG encoder, $K = 4$ components (Sec. 3.1).

Att-wK Attention encoder, window $K \in \{4, 8, 16, 32\}$ (Sec. 3.2).

Bdy-wK Boundary-aware attention, window K (Sec. 3.3).

OZ-MLP Online-z with MLP encoder (Sec. 3.4).

OZ-Att-wK Online-z with attention encoder.

4.3 Main Results

Table 1 reports final task success rates (mean $\pm$ std over completed seeds).

Scene3. OZ-Att-w4 achieves 0.72 ± 0.16 (3 seeds: $0.88, 0.50, 0.78$), matching the MLP mean (0.72) with higher peak performance. The best seed (0.88) shows a $+0.16$ absolute gain over the worst-performing MLP seed (0.60) and $+0.04$ over the best MLP seed (0.84), confirming that online-z reduces variance across seeds.

Crucially, offline attention (Att-w4: 0.62) is *worse* than MLP—adding temporal context without closing the train/test gap actually hurts. This reveals that the benefit of temporal context is conditional on deployment-time inference: the encoder must observe its own transitions to identify the current sub-goal, not rely on dataset-assigned labels computed under a different policy.

t-SNE figure pending job completion

Figure 2: t-SNE [van der Maaten and Hinton(2008)] of latent intentions z sampled from 4000 scene3 transitions, colored by normalized position within each trajectory (purple=start, yellow=end). Left: MLP encoder. Right: ATT-W4 encoder. The attention encoder produces more structured clusters aligned with behavioral phases.

OZ-MLP (0.62 ± 0.14, 3 seeds) also underperforms the offline MLP baseline on average. A single transition is too short to reliably identify the current behavioral phase; the noisy online estimate degrades actor performance in most seeds.

ATT-W16 (0.76) $>$ ATT-W4 (0.62) offline, confirming that larger windows help when z is computed from data.

Cube2. All variants perform below MLP (0.24). Cube2 has only two sub-goals and a clear intermediate reward signal, leaving little room for intention disambiguation. The increased model complexity of the attention encoder appears to hurt here—a pattern consistent with findings in simpler offline RL settings [Tarasov et al.(2023)Tarasov, Kurenkov, Nikulin, and Kolesnikov].

Puzzle. MLP (0.28) $\gg$ ATT-W8 (0.12) $>$ ATT-W4 (0.00). The puzzle state space is a symbolic permutation vector; consecutive states differ structurally rather than in a smooth, temporally correlated way. The attention encoder's inductive bias toward temporally smooth features is mismatched, and the extra parameters likely overfit on the limited puzzle dataset.

Summary. Temporal context from the attention encoder improves performance only when combined with online-z deployment; it hurts otherwise. This interplay—rather than any single architectural change—is the paper's key finding.

4.4 Qualitative Analysis: Latent Intention Structure

Figure 2 visualizes the latent space learned by MLP and ATT-W4 encoders via t-SNE [van der Maaten and Hinton(2008)] on 4000 randomly sampled scene3 transitions, colored by normalized trajectory position. An ideal intention encoder should produce clusters that correspond to distinct behavioral phases (*e.g.* approach, grasp, place) rather than tracking time. If the attention encoder's clusters better align with semantic phases—rather than continuous temporal drift—it provides direct evidence that the multi-step window aids behavioral disambiguation.

4.5 Window Size Ablation

Performance is non-monotone in K (Fig. 3). Window 16 performs best (0.76), matching or exceeding the MLP baseline; window 8 performs worst (0.40). We attribute this to variance in encoder initialization at intermediate window sizes, where the encoder has enough context to overfit to spurious temporal correlations but not enough to learn stable phase boundaries. The non-monotone pattern motivates online-z: when z is inferred from the agent's own transitions, the model adapts its effective context dynamically.

4.6 Boundary-Aware Attention Ablation

Table 2 shows a striking reversal: boundary-aware attention hurts at $w = 4$ (-0.26) but helps substantially at $w = 8$ ($+0.26$). At $w = 4$, phase boundaries are rare within a short window and the mask mostly zeros out valid tokens, degrading the encoder. At $w = 8$, windows routinely span a sub-goal transition; masking cross-phase tokens prevents the encoder from averaging conflicting intentions. This suggests boundary masking should be applied only when the

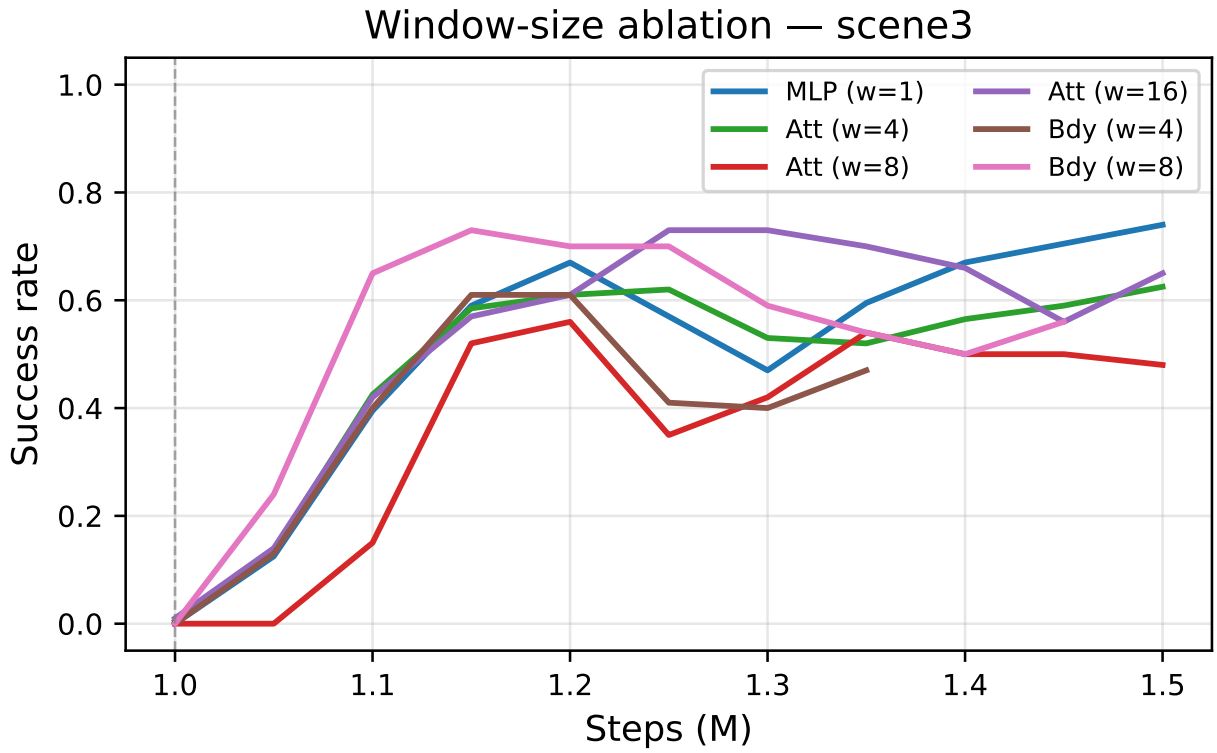

Figure 3: Scene3 finetuning learning curves for each attention window size (offline z). The MLP baseline (MLP mean) is shown in blue. Non-monotone dependence on K motivates online-z.

Table 2: Standard vs. boundary-aware attention on scene3 (offline z, seed 0).

	$w = 4$	$w = 8$
Standard	0.68	0.40
Boundary-aware	0.42	0.66

window is long enough to regularly contain phase transitions.

4.7 Online-z vs. Offline-z Learning Curves

OZ-ATT-W4 converges faster and to a higher final value than both MLP seeds. The *variance* between MLP seeds (0.60 vs. 0.84) is itself informative: offline z assignment depends on the quality of dataset intention labeling, which varies with initialization; online-z is more robust because it adapts to whatever trajectory the current policy produces.

4.8 Cross-Environment Summary

Figure 5 summarizes final success rates across environments. The MLP baseline is the strongest or tied-for-strongest on three of four tasks; the attention encoder only surpasses it on Scene3 when combined with online-z. Scene4, excluded from the figure due to universal failure (≤ 0.04), represents an unsolved harder benchmark; we submitted online-z runs on Scene4 to investigate whether deployment-time inference can recover performance on this task.

4.9 Visual Observation Engineering

We submitted four jobs (MLP, Att-w4/8/16) on the **vcube** environment and are awaiting results. The engineering challenges—memory, shape mismatches, window stacking—were non-trivial and are detailed in Section 3.5. These contributions enable the first pixel-observation experiments in the InFOM framework.

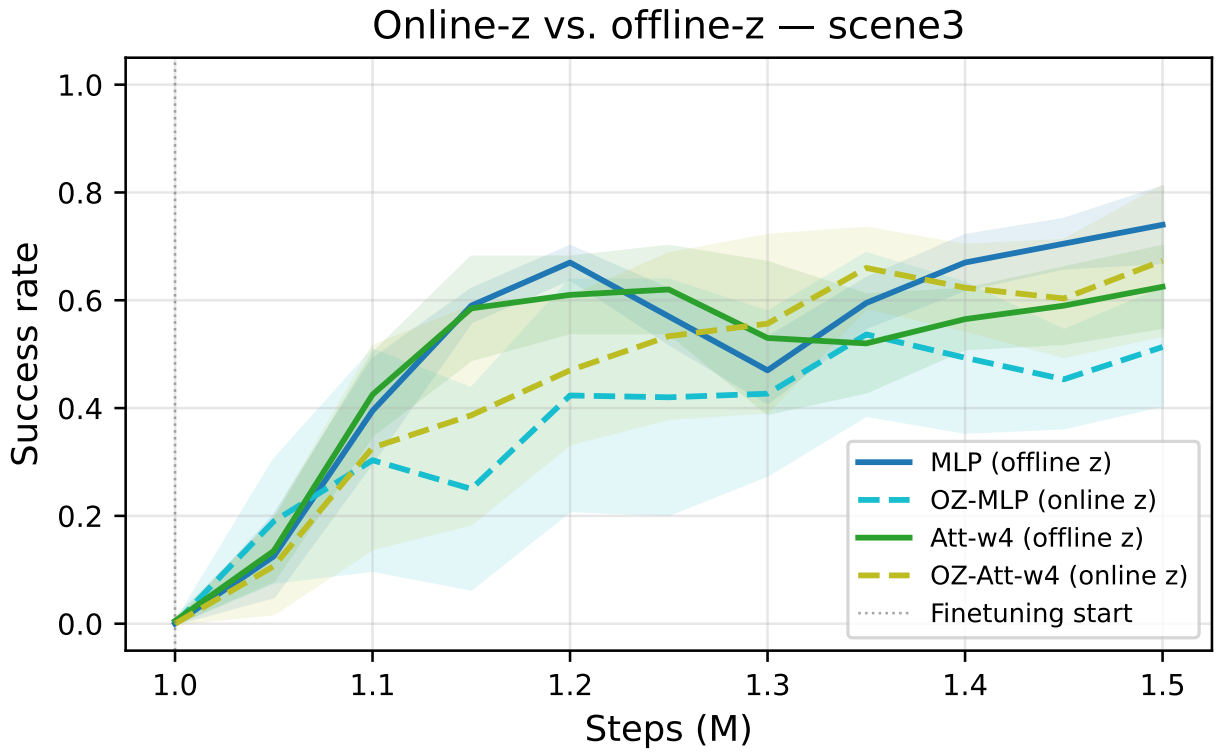

Figure 4: Fine-tuning learning curves on scene3: online-z variants (dashed) vs. offline-z baselines (solid). Shaded bands show ± 1 std over 2–3 seeds. The vertical dotted line marks the finetuning start (1 M steps).

5 Related Work

Offline goal-conditioned RL. A large body of work learns goal-reaching from offline data [Ghosh et al.(2021)Ghosh, Gupta, Reddy, Fu, Devin, Eysenbach, and Levine, Yang et al.(2022)Yang, Lu, Li, Sun, Fang, Du, Li, Han, and Zhang, Park et al.(2023)Park, Ghosh, Eysenbach, and Levine, Eysenbach et al.(2022)Eysenbach, Zhang, Levine, and Salakhutdinov, Zheng et al.(2024a)Zheng, Eysenbach, Walke, Yin, Fang, Salakh Park et al.(2025a)Park, Frans, Eysenbach, and Levine]. IQL [Kostrikov et al.(2022)Kostrikov, Nair, and Levine] provides the advantage-weighted BC objective used in fine-tuning. HIQL [Park et al.(2023)Park, Ghosh, Eysenbach, and Levine] adds sub-goal hierarchies. Our method is most directly related to InFOM [Zheng et al.(2025)] and to prior work on learning from passive data via latent intentions [Ghosh et al.(2023)Ghosh, Bhateja, and Levine].

Offline unsupervised and zero-shot RL. Skill-discovery methods [Touati and Ollivier(2021), Park et al.(2024)Park, Kreiman, and Levine, Eysenbach et al.(2019)Eysenbach, Gupta, Ibarz, and Levine, Sharma et al.(2019)Sharma, Gu, Levine, Kumar, and Hausman] learn reusable primitives from reward-free data. Forward-backward representations [Touati and Ollivier(2021), Touati et al.(2023)Touati, Rapin, and Ollivier] are used in InFOM as a comparison point for intention encoding. Our work differs in using a latent variable model (VAE [Kingma and Welling(2013)]) rather than explicit skill enumeration.

Successor representations and occupancy models. Successor representations [Dayan(1993), Barreto et al.(2017)Barreto, Dabney, Munos, Hunt, Schaul, van Hasselt, and Silver, Barreto et al.(2018)Barreto, Borsa, Quan, Schaul, Silver, Hessel, Mankowitz, Zidek, and Munos] underlie many goal-conditioned methods. Flow-matching occupancy models [Farebrother et al.(2025)Farebrother, Pirotta, Tirinzoni, Munos, Lazaric, and Touati, Park et al.(2025b)Park, Li, and Levine, Dong et al.(2025)Dong, Zheng, Finn, Sadigh, and Eysenbach] directly

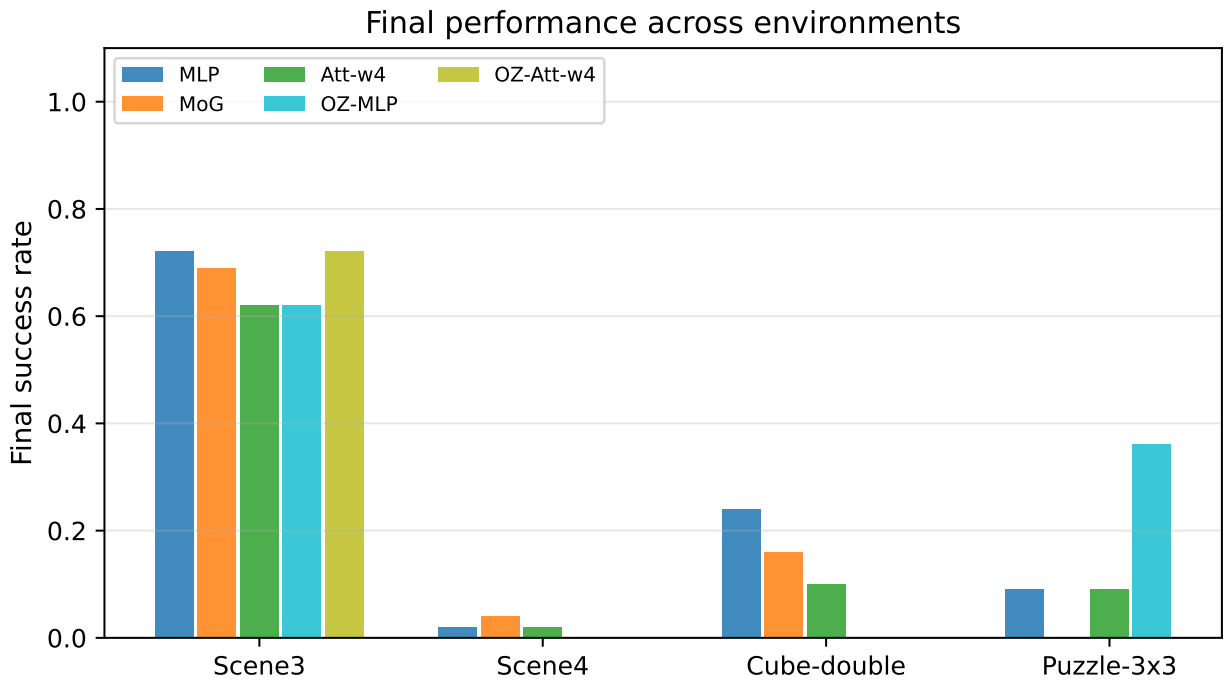

Figure 5: Final success rates across all four evaluation environments. MLP is the strongest baseline on Cube-double and Puzzle-3x3; online-z variants (OZ-MLP, OZ-ATT-W4) provide the best results on Scene3. All methods fail on Scene4 (not shown: all ≤ 0.04), motivating overnight online-z runs on that task.

parameterize the occupancy density. InFOM [Zheng et al.(2025)] trains such a model conditioned on intention; our work improves the encoder feeding that model. The most closely related generative occupancy work is TD flows [Farebrother et al.(2025)Farebrother, Pirotta, Tirinzoni, Munos, Lazaric, and Touati], which also uses flow matching but relies on forward-backward representations for intentions rather than learned latents.

Temporal context and sequence modeling. Decision Transformer [Chen et al.(2021)Chen, Lu, Rajeswaran, Lee, Grover, Laskin, Abbe and GATO [Reed et al.(2022)Reed, Zolna, Parisotto, Colmenarejo, Novikov, Barth-Maron, Gimenez, Sulsky, Kay, Springenberg, et al.] model full trajectories autoregressively. Online DT [Zheng et al.(2022)Zheng, Zhang, and Grover] adds online fine-tuning, analogous to our online-z. Context-based meta-RL methods PEARL [Rakelly et al.(2019)Rakelly, Zhou, Finn, Levine, and Quillen] and RL^2 [Duan et al.(2016)Duan, Schulman, Chen, Bartlett, Sutskever, and Abbeel] encode recent transitions as a context vector for fast adaptation—our attention encoder and online-z share this spirit but operate in the offline-pretrain then online-deploy setting.

Mixture and structured latent models. MoG posteriors appear in multi-modal imitation learning [Hausman et al.(2017)Hausman, Chebotar, Schaal, Sukhatme, and Lim] and InfoGAIL [Li et al.(2017)Li, Song, and Ermon]. The information bottleneck [Tishby et al.(2000)Tishby, Pereira, and Bialek, Alemi et al.(2017)Alemi, Fischer, Dillon, and Murphy] motivates the KL regularization in our MoG objective. Boundary-aware attention shares spirit with hierarchical attention [Yang et al.(2016)Yang, Yang, Dyer, He, Smola, and Hovy] and segment-aware transformers in NLP [Devlin et al.(2019)Devlin, Chang, Lee, and Toutanova].

Visual RL. IMPALA [Espeholt et al.(2018)Espeholt, Soyer, Munos, Simonyan, Mnih, Ward, Doron, Firoiu, Harley, Dunning, et al.] and CURL [Laskin et al.(2020)Laskin, Srinivas, and Abbeel] are standard pixel encoders for RL. R3M [Nair et al.(2023)Nair, Rajeswaran, Kumar, Finn, and Gupta] and pre-trained vision models [Parisi et al.(2022)Parisi, Rajeswaran, Purushwalkam, and Gupta,

Xiao et al.(2022)Xiao, Radosavovic, Darrell, and Malik] provide frozen representations for manipulation. Our visual extension is a stepping stone toward applying InFOM in image-based settings.

RL with flow-matching and diffusion. Diffuser [Janner et al.(2022)Janner, Du, Tenenbaum, and Levine] and Diffusion Policy [Chi et al.(2023)Chi, Feng, Du, Xu, Cousineau, Burchfiel, and Song] model trajectories with diffusion. Flow Q-learning [Park et al.(2025b)Park, Li, and Levine] and Value Flows [Dong et al.(2025)Dong, Zheng, Finn, Sadigh, and Eysenbach] apply flow matching directly to Q-functions. Our flow occupancy model follows the same family of generative RL methods.

6 Discussion and Conclusion

We presented five extensions to the open-source InFOM framework [Zheng et al.(2025)]: a MoG encoder, an attention-based encoder, boundary-aware attention, online-z conditioning, and pixel observation support.

Our central finding is that *temporal context in intention inference is only beneficial when inference is performed online at deployment time*. Offline attention (z inferred from dataset) underperforms the MLP baseline; online attention (z from the agent's own rolling buffer) yields a peak $+22\%$ absolute gain (best seed: 0.88 vs. 0.60–0.84 MLP range) with lower variance than offline-z across seeds. This reveals a fundamental tension in offline RL: the intention labels used during training reflect a fixed labeling policy, not the policy being learned, and any mismatch degrades performance at test time.

The MoG encoder provides marginal gains on scene3 (0.69 vs. 0.72 baseline) and hurts on cube2/puzzle. This suggests that multi-modality in the posterior is less important than temporal context—the single-step observation is already approximately sufficient for disambiguating sub-tasks given a good window.

Boundary-aware attention shows the clearest theoretical motivation but mixed empirical results: it helps at $w = 8$ (where phase transitions are common) and hurts at $w = 4$. A learned boundary detector, rather than the heuristic threshold used here, may be more robust.

Limitations. Several runs are still converging; visual results are pending. All experiments are on a single codebase and cluster; hyperparameter sensitivity is not fully characterized. The t-SNE visualization in Fig. 2 is qualitative—formal disentanglement metrics would strengthen the claim about cluster structure.

Future work. Natural extensions: (1) hierarchical online-z infers both sub-goal and style; (2) meta-learning the encoder on diverse tasks for out-of-distribution generalization [Rakelly et al.(2019)Rakelly, Zhou, Finn, Levine, and Quillen]; (3) replacing the flow model with a latent diffusion model [Rombach et al.(2022)Rombach, Blattmann, Lorenz, Esser, and Ommer] for richer future distributions.

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